Panskura Banamali College (Autonomous)

B.Sc.(H) 4th Semester Examination -2021 (Online Mode)

Subject: Computer Science **Paper: CC8P**

Paper Name: Design and Analysis of Algorithms Lab

F.M.: 20 **Time : 2 Hours**

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2020UG009967, paper_code : CC8P , then file name will be :: 2020UG009967- CC8P.pdf for question 'A' and 2020UG009967- CC8P.mp3 for question 'B']

A. Answer any one question: **1 x12 = 12**

1. Write the algorithm to implement stack using C .
2. Write the algorithm to implement Breadth-First Search in a graph.
3. Write the algorithm to implement Depth-First Search in a graph.
4. Write the algorithm to determine the minimum spanning tree of a graph.
5. Write the algorithm to implement Heap sort (The program should report the number of comparisons).
6. Write the algorithm to implement Heap Sort(The program should report the number of comparisons)
7. Write the algorithm to implement Randomized Quick sort (The program should report the number of comparisons).
8. Write the algorithm to implement Merge Sort(The program should report the number of comparisons)

Marks distribution:

a. Problem Definition - 1 Marks
b. Algorithm/Flow Chart/ Pseudo-code– 5 Marks
c. Program listing - 3 Marks
d. Output/Result with discussion- 3 Marks

B. Answer any four questions: **4 x2 = 8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is an algorithm?
2. What is time complexity of linear Search?
3. What is the need for an algorithm?
4. What is the Complexity of Algorithm?
5. What are the Asymptotic Notations?
6. Give some examples of Divide and Conquer algorithm?
7. What is a binary search?
8. What Are The Criteria Of Algorithm Analysis?
9. What are the differences between stack and Queue?
10. What is the difference between the Singly Linked List and Doubly Linked List data structure?

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